

CONSTRUCTION OS

PRD - Procurement Dashboard

Procurement Manager Role Workspace

Product purpose

Define the exact Procurement Manager role shell, fixed dashboard, sourcing and purchasing control surfaces, approval projections and cross-module behavior. The experience is exception-first: know what demand is open, what sourcing is blocked, what is committed, what is late and what requires a decision.

Field	Value
Document	PRD_Procurement_Dashboard
Version	1.0
Status	Implementation-ready role dashboard specification
Primary roles	Procurement Director / Manager; compatible with scoped Procurement Manager templates
Base requirements	PRD_Procurement_Module + Complete Role & Permission Matrix + Approvals & Workflow Engine
Connected domains	Projects, Inventory, Assets, Finance, Contracts, Documents, Tasks, Calendar, Notifications, Workflow
Scope	Procurement role dashboard + exact sidebar + management workspaces and source-linked controls
Date	2026-08-07

Permanent dashboard scope lock

The Procurement Dashboard and Procurement role sidebar shall contain only the elements explicitly defined in this PRD. Developers must not add, infer, duplicate, substitute or auto-expose extra dashboard widgets, modules or shortcuts without an approved PRD revision. Platform Configuration may hide approved elements; it may not create unapproved ones.

1. Document Authority, Precedence and Product Position

Priority	Authority	Controls
1	Complete Role & Permission Matrix	Procurement role authority, capability scope, protected commercial fields, separation of duties, deny-by-default behavior and revocation.
2	PRD_Procurement_Module v1.0	Supplier purchasing profile, demand, packages, RFQs, quotations, comparison, awards, POs, delivery, supplier performance, APIs and dashboard source metrics.
3	Approvals & Workflow Engine v1.0	Decision routing, approval matrices, participants, work items, delegation, SLA, SoD and source finalization.
4	Role Context and Shared Module Navigation	Procurement shell remains active when authorized shared Project, Inventory, Finance, Document, Task, Calendar or Workflow content opens.
5	Platform UI/UX Architecture	Shared shell, Role Center, Work Inbox, Quick Create, progressive disclosure, responsive behavior, accessibility and motion.
6	Connected source PRDs	Inventory, Finance, Contracts, Assets, Projects, Documents, Tasks and Calendar retain authority for their own records.
7	This PRD	Exact Procurement Manager dashboard composition, sidebar, workspace behavior and role-specific UX decisions.

1.1 Precedence note - approvals are no longer deferred

Resolved dependency

The Procurement Module was authored before the final Approvals & Workflow Engine and therefore describes central approval routing as future scope. The Workflow PRD now exists and governs routing, participants, decisions, delegation, SLA and escalation. Procurement still owns the purchasing record and final business state. This dashboard integrates Workflow as a shared decision surface without moving procurement ownership into Workflow.

1.2 Executive purpose

- Give Procurement Managers one calm operating surface across demand, sourcing, commitments, deliveries, suppliers and commercial exceptions.
- Prioritize actionable exceptions and decisions before passive analytics.
- Keep company, project, category, buyer/team and period context explicit.
- Make common procurement work reachable within two navigation decisions while preserving full auditability.
- Prevent dashboard convenience from bypassing issued-record immutability, approval, quantity allocation, budget, confidentiality or cross-module ownership rules.

Final product position

Procurement Manager opens Dashboard -> sees demand and sourcing pressure -> sees orders and deliveries at risk -> resolves decisions/exceptions -> opens the authoritative procurement record -> completes the controlled action. The dashboard is a permission-safe operational projection, not a second procurement database.

2. Goals, Non-Goals and Role Scope

In scope	Out of scope
Exact Procurement role shell, sidebar and fixed dashboard	A separate purchasing application that copies Finance, Inventory, Contracts, Assets or Project records
Company/project/category/buyer scoped procurement portfolio	Automatic visibility of every group company or confidential procurement process
Demand, sourcing, ordering, delivery, supplier and commercial summaries	Arbitrary dashboard drag-and-drop or user-added widgets in v1
My Work / Work Inbox and Workflow approval projections	Workflow ownership of Purchase Requests, RFQs, awards, POs or delivery records
Supplier qualification, performance and risk projections	Supplier Portal, e-auctions, OCR, AI sourcing or external risk services in initial scope
Permission-safe requested/quoted/committed/received/invoiced context	General ledger, AP posting, payment execution or statutory accounting
Responsive desktop/tablet/mobile web	Full offline synchronization or live courier/EDI/marketplace integrations in v1

2.1 Role scope model

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AUTHENTICATED USER
-> Active Company / Group Scope
-> Active Role: Procurement Manager
-> Authorized Procurement Scope
    -> Optional Project / Category / Buyer / Period Filters
    -> Procurement Dashboard / Shared Source Pages
  
```

- Procurement authority is scoped by tenant, active company/group, project, category, record and capability.
- Filters narrow authorized data; they never expand access.
- Competing quotations, evaluation notes, negotiation limits and supplier bank changes are independently protected fields/actions.
- Multi-role users remain in the active Procurement role until they explicitly switch roles.

3. Application Shell, Header and Exact Sidebar

3.1 Persistent header

```
PROCUREMENT
[ Company v ] [ All Projects v ] [ Category v ] [ Buyer / Team v ] [ Period v ]
[ Global Search ] [ + New ] [ Notifications ] [ Profile / Role ]
```

- Company/group changes re-resolve authorization, dashboard scope, saved views and sidebar visibility.
- Project, category and buyer/team are explicit scope filters; period controls date-based summaries only.
- Global Search is permission safe and opens results inside the active Procurement shell.
- Notifications deep-link to source records or workflow work items without changing active role.

3.2 Exact v1 sidebar

```
PROCUREMENT
  Dashboard
  My Work
  Exceptions

DEMAND
  Purchase Requests
  Replenishment Requests
  Emergency Requests

PACKAGES
  Procurement Packages

SOURCING
  RFQs
  Quotations
  Supplier Comparison
  Negotiations
  Award Recommendations

ORDERING
  Purchase Orders
  Framework / Call-Off Orders

DELIVERY
  Delivery Schedule
  Expected Receipts
  Service Receipts
  Returns & Claims

SUPPLIERS
  Supplier Directory
  Qualification
  Performance & Risk

WORK
  Approvals
  Tasks
  Calendar
  Documents

REPORTING
  Reports
  Analytics

SETTINGS
  Procurement Settings
```

- Group labels are visual separators, not routes.
- Unauthorized or configuration-disabled items disappear with clean reflow.
- Approvals opens the shared Workflow surface inside Procurement context.
- Inventory receipts, Finance invoices/payments, Contracts, Projects and Assets are linked content, not extra Procurement sidebar modules.

4. Full Procurement Dashboard Architecture Map



Layout lock

These dashboard regions are the complete v1 Procurement Dashboard. There is no generic company KPI area, unrestricted Finance panel, hidden employee scoring area, arbitrary chart zone or user-added widget region. Deeper analytics live in Reports/Analytics and authoritative source modules.

5. Primary Cues - Exact Set

Cue	Definition	Click-through
Open Purchase Requests	Authorized purchase demand not completed/cancelled, including submitted requests awaiting sourcing or further controlled action.	Purchase Requests register with same scope.
Open RFQs	Issued/active authorized RFQs that have not reached final closed/cancelled source state.	RFQs register filtered to active/open.
Open Purchase Orders	Authorized POs with remaining commercial or delivery obligation according to source state.	Purchase Orders register with open quantity/status.
Deliveries At Risk	Authorized deliveries delayed, partial, rejected, overdue or otherwise source-flagged as quantity/date risk.	Delivery Exceptions / Schedule filtered to risk.
Supplier Exceptions	Authorized supplier qualification/document/status/performance/risk exceptions requiring attention.	Supplier Performance & Risk filtered to exception.
My Pending Actions	Assigned Procurement work items: requests, evaluations, negotiations, order actions, delivery exceptions, supplier reviews and approvals/tasks.	My Work / Work Inbox.

- Each cue displays a current count/status and optional governed value where explicitly supported.
- Counts reconcile exactly with drill-down registers using the same tenant/company/project/category/buyer/date and permission scope.
- Dashboard values with money state currency scope and date basis; no unexplained amount is permitted.
- Cards use cached/precomputed summaries; drill-down opens authoritative live records.

6. My Work / Needs My Attention

Work group	Examples	Source / action
Demand	Overdue request, emergency demand, missing specification, request returned for correction.	Purchase Request / Package.
Sourcing	RFQ response due, clarification unresolved, quotation validity expiring, evaluation overdue.	RFQ / Quotation / Comparison.
Award	Evaluation assigned, recommendation pending, approval/information request, conflict-of-interest intervention.	Comparison / Award / Workflow.
Ordering	PO ready to issue, PO unacknowledged, revision pending, framework capacity exception.	Purchase Order / Framework.
Delivery	Delayed/partial/rejected delivery, shortage/damage, service receipt exception, return/claim.	Delivery / Receipt link / Return / Claim.
Supplier	Qualification/document expiry, suspension/risk review, performance deterioration.	Supplier record.
Execution	Linked task, comment/correction, deadline or meeting action.	Tasks / Calendar / source record.

6.1 Priority ordering

- Critical project delivery risk, emergency procurement and blocked high-value decisions appear before routine work.
- Overdue and SLA-escalated workflow items appear before future-due work.
- Commercial confidentiality is preserved: users see only work they are allowed to know exists.
- Work Inbox status is a projection. Closing or updating the source/Workflow record removes the item after confirmed refresh.

6.2 Default views

My Priority		Due Today		Overdue		Waiting on Others		Escalated
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7. Procurement Pipeline

Demand -> Sourcing -> Comparison -> Negotiation -> Ready to Order -> Ordered -> Delivery -> Completed

7.1 Pipeline purpose

- Pipeline is a management projection over Purchase Requests, Procurement Packages and configured procurement cases.
- Board, table and timeline views must resolve status from the same backend domain state.
- Dragging/proposing a stage change only initiates a valid source transition; it never bypasses required fields, locks, approvals or validation.
- Exception lane includes overdue, missing specification, supplier-document expiry, budget gap, unacknowledged order and delayed delivery.

7.2 Card content

Field group	Required card content
Identity	Code, title/package name, record type.
Scope	Company/project, category/work package where relevant.
Commercial	Authorized value/currency or amount-at-risk where source permits.
Responsibility	Responsible buyer/team and current stage.
Time	Required date, next deadline or delivery milestone.
Risk / action	Source risk indicator and next valid action.

No client-only status

The pipeline cannot maintain a private client-side lifecycle. Server state is authoritative, and transition conflicts must reload the current source version.

8. Demand and Purchase Requests

8.1 Dashboard Demand panel

Metric / list	Definition
Open Requests	Authorized requests not completed/cancelled.
Overdue Requests	Required date or governed due date passed while unresolved.
Emergency Requests	Requests using configured emergency classification/override path.
Value Awaiting Sourcing	Permission-safe value of eligible demand not yet committed to sourcing/order.
By Project / Company / Category	Distribution using exact source dimensions.

8.2 Purchase Request behavior

- One request may contain stock items, assets, services, rentals, subcontract work or mixed lines.
- Lines may carry project, work package, cost center, warehouse, delivery location and required date.
- Submitted requests store immutable snapshots; revisions create controlled new versions.
- Duplicate-demand warnings and replenishment-origin links remain source module behavior.
- Request status and amount must drill to exact lines and history; dashboard cannot collapse materially different source states into a false single lifecycle.

8.3 Request actions

Create Draft -> Validate -> Submit -> Workflow / Authorization Checkpoint -> Source Confirmed -> Package / RFQ / Direct Order path

9. Sourcing, RFQs and Quotations

9.1 Dashboard Sourcing panel

Metric / list	Definition
Open RFQs	Active issued RFQs.
Supplier Response Rate	Governed ratio of invited suppliers with valid response state.
Quotations Due	RFQ/quotation deadlines approaching or due.
Evaluations Overdue	Assigned technical/commercial evaluations past due.
Average RFQ Cycle Time	Governed sourcing metric with explicit date basis.
Sourcing Pipeline	RFQs by active source stage.

9.2 RFQ controls

- RFQ requires valid deadline, at least one line and at least one invited supplier.
- Issued RFQ revision is immutable; changes use a new controlled revision/cancellation path.
- Clarifications are linked to exact RFQ revision and audience scope.
- Supplier invitation visibility respects qualification/status restrictions and commercial confidentiality.
- RFQ issue/release may be routed through Workflow when configured.

9.3 Quotation integrity

- Original and revised supplier quotations remain linked; a new revision never silently replaces the prior offer.
- Totals reconcile to lines, discounts, taxes and charges.
- Competing quotations and sensitive pricing use field/record permissions and sensitive-read audit where configured.

10. Supplier Comparison, Evaluation and Negotiation

10.1 Comparison workspace

SCOPE / LINE SELECTOR			
Supplier A	Supplier B	Supplier C	...
Price	Price	Price	
Lead Time	Lead Time	Lead Time	
Compliance	Compliance	Compliance	
Quality	Quality	Quality	
HSE	HSE	HSE	
Warranty	Warranty	Warranty	
Performance	Performance	Performance	
Risk	Risk	Risk	

Normalized total / evaluation / recommendation			

10.2 Required behavior

- Comparison stores a versioned snapshot and normalization rules.
- Technical and commercial evaluations may be separate, parallel or combined according to configured process.
- Evaluation weights total 100% when weighted scoring is enabled.
- Every score, override and recommendation stores evaluator, before/after values, reason and source revision.
- Large comparison matrices are virtualized; mobile renders supplier-by-supplier cards instead of shrinking the desktop matrix.
- Negotiation rounds preserve history; negotiation limits/notes are sensitive commercial fields.

10.3 Separation of duties

Award preparation is not sole approval

A Buyer or Procurement Manager may prepare/evaluate/recommend according to capability, but separation-of-duty policy can prevent the same actor from being the sole final approver of an award. Workflow resolves exact participants and conflicts.

11. Award Recommendations and Workflow Decisions

11.1 Award lifecycle

Comparison -> Evaluation Complete -> Negotiation (optional) -> Award Recommendation -> Workflow Decision
-> Source Award Selected / Returned / Rejected

11.2 Workflow boundary

Workflow owns	Procurement owns
Routing, participant resolution, work items, decisions, delegation, SLA and escalation.	Award recommendation content, source version, eligible quantities, supplier selection result and final Procurement status.
Decision completion and source-finalization request.	Final business transition and side effects after validation.
Workflow audit of selection/routing/decision.	Record-level procurement activity and issued commercial history.

- Approval and source finalization are separate. A workflow may be approved while Procurement is still Source Finalization Pending because of a version conflict or business validation failure.
- Workflow participants receive only the record content required by policy and exact source version.
- Conflict declarations, delegation and substitutes are recorded without silently skipping mandatory participants.
- Emergency procurement may use explicit controlled override/bypass rules only when configured; there is no generic Skip Approval action.

12. Purchase Orders and Commercial Control

12.1 Dashboard Ordering panel

Metric / list	Definition
Open POs	Issued/active POs with remaining open commercial or delivery obligation.
Committed Value	Permission-safe committed value from Procurement PO/Finance commitment reference.
Unacknowledged POs	Issued POs awaiting required supplier acknowledgment.
Overdue PO Lines	PO lines past governed required/expected date while open.
Revised Orders	Orders with controlled revision history requiring visibility.
Framework Capacity	Remaining/expiring framework ceiling/effective-date exposure where applicable.

12.2 Purchase Order rules

- PO may be created from approved award or explicit direct-order flow when permitted.
- Supplier, currency and company must match selected award unless the direct-order path explicitly applies.
- PO totals reconcile before issue.
- Issued PO revisions are immutable; changes use a new revision, cancellation of remaining quantity or controlled change process.
- Issue/revision may trigger Finance commitment events but does not post accounting entries.
- Supplier acknowledgment is captured as a source-linked action; Supplier Portal participation is optional/future unless separately implemented.

12.3 Approval matrix example integration

Workflow matrix

The Workflow Engine supports value-based Procurement routing, including purchase-order thresholds and Procurement/Finance/Executive participants. Exact matrices are company configuration, not hard-coded dashboard logic.

13. Delivery, Receiving, Returns and Claims

13.1 Dashboard Delivery panel

Metric / list	Definition
Due This Week	Authorized delivery schedule segments due inside the selected period.
Delayed	Supplier delivery has passed expected date or source indicates delay.
Partial	Delivery/receipt state indicates partial fulfillment.
Rejected	Source-linked rejected delivery/receipt quantity or service acceptance.
In Transit	Procurement delivery state indicates dispatched/in transit.
Quantity At Risk	Open quantity affected by delay, shortage, rejection or source exception.
Critical Project Deliveries	Delivery exceptions flagged critical by governed project/procurement rules.

13.2 Ownership rules

- Procurement owns delivery expectation, supplier shipment/arrival context and commercial exception record.
- Inventory owns authoritative physical goods receipt and posted stock movements.
- Procurement displays exact linked Inventory Receipt summary/status and never fabricates a goods receipt.
- Service/subcontract receipt is supported without fake Inventory movement; source records enforce accepted quantity/value tolerance.
- Return quantity cannot exceed accepted quantity net of prior returns.
- Claims preserve links to delivery, receipt, supplier performance and Finance consequences.

13.3 Delivery exception flow

P0 / Delivery Schedule -> Dispatch -> Arrival -> Inventory Receipt or Service Receipt -> Exception? -> Shortage / Reject / Return / Claim -> Resolve -> Reconcile

14. Supplier Health, Qualification and Risk

14.1 Dashboard Supplier Health panel

Metric / list	Definition
Qualified	Suppliers currently qualified in the visible procurement scope.
Expiring Documents	Required supplier documents/qualification validity approaching expiry.
Suspended	Suppliers in controlled suspended state.
Concentration Risk	Governed single-source/concentration risk flag where calculated/configured.
Top Performers	Only governed supplier performance ranking/metric with visible methodology.
Performance Deterioration	Source score/trend crossing a configured deterioration rule.

14.2 Supplier Page architecture

Supplier

+-- Overview

+-- Qualification

+-- Categories

+-- Contacts

+-- Documents

+-- RFQs / Quotations

+-- Purchase Orders

+-- Deliveries / Returns / Claims

+-- Performance

+-- Risk

+-- Contracts / Finance links (permission-safe)

`-- Activity

14.3 Supplier lifecycle

Prospect -> Under Qualification -> Qualified -> Preferred -> Suspended -> Blacklisted -> Archived

Suspended/Blacklisted suppliers remain historically visible but new invitations/orders are blocked or require configured exception with reason.

Supplier bank references, competing pricing and sensitive risk/negotiation content use independent field policy.

Shared legal/company identity is linked from master data; Procurement does not duplicate group/company identity.

15. Commercial Summary

15.1 Exact dashboard commercial region

Metric	Rule
Estimated Savings	Only when a governed baseline and calculation method exist; currency/date basis must be explicit.
Price Variance	Source-defined comparison of normalized price/reference basis with transparent definition.
Off-Contract Spend	Only where contract/framework matching rules are implemented and governed.
Budget Exposure	Permission-safe Finance budget/commitment context; Finance remains authority.
Committed vs Received vs Invoiced	Cross-domain projection with each source state labeled and linked.

15.2 Cross-source status vocabulary

Requested (Procurement)
Quoted (Procurement)
Committed (Procurement + Finance reference)
Received (Inventory / Service Receipt)
Invoiced (Finance)
Paid (Finance)

- The dashboard must never present a Finance invoice/payment value as if Procurement owns it.
- Currency normalization stores rate/source/date used for comparisons where relevant.
- Estimated and source-final values are visually distinguished.
- Every amount drills to exact records and preserves current company/project/currency scope.

16. + New and Quick Actions

16.1 + New menu - exact allowed entries

Action	Availability / rule
Purchase Request	Shown with purchase_requests.create. Draft does not bypass submission validation.
Procurement Package	Shown with procurement_packages.manage and valid company scope.
RFQ	Shown with rfqs.create; issue remains a separate controlled action.
Supplier	Shown with suppliers.manage; qualification/status actions remain separate.
Purchase Order	Shown only where purchase_orders.create and an award/direct-order context is valid.
Delivery Exception	Shown with deliveries.manage and valid PO/delivery source.
Task	Shared Task create shown only with Tasks capability and procurement context.

16.2 Contextual quick actions

- Convert Request Lines to RFQ starts from selected Purchase Request lines.
- Create Comparison starts from valid quotation responses.
- Recommend Award starts from a valid comparison/evaluation state.
- Issue PO is never a generic dashboard action without source validation/Workflow where required.
- Record Delivery Exception requires a linked Delivery/PO context.
- High-risk actions are server-confirmed and never optimistic.

Quick Create rule

Quick Create starts a valid draft or source-specific action. It cannot skip required company/project dimensions, commercial validation, issued-record immutability, quantity locks, approvals or separation-of-duty controls.

17. Procurement Workspace - Board, Table and Timeline

17.1 Purpose

- The Procurement Workspace connects demand to fulfillment across Purchase Requests, Packages and configured procurement cases.
- It is a visual operating board, not a second source of status.
- Board, Table and Timeline are alternate projections over the same source records.
- Saved views, group-by, filters, column chooser, Personal Stars and compact/detailed modes are allowed source-module features.

17.2 Workspace lanes

Demand | Sourcing | Comparison | Negotiation | Ready to Order | Ordered | Delivery | Completed
Exception lane: Overdue | Missing Specification | Supplier Expiry | Budget Gap | Unacknowledged PO | Delayed Delivery

17.3 Manager workload view

- Workload indicators by buyer/team may show assigned record counts, due work and queue distribution.
- No hidden productivity score, surveillance metric or subjective employee ranking is permitted.
- Buyer filter and workload view respect role/team scope and commercial confidentiality.

18. Procurement Package Workspace

18.1 Role of the Package

End-to-end commercial container

A Procurement Package groups coordinated demand, sourcing, orders, contracts, deliveries and commercial context around a project/company procurement scope such as Structure, Facade, MEP, Fit-Out or Site Logistics. It links records; it does not replace them.

18.2 Exact workspace tabs

Overview

Demand

RFQs

Quotations

Comparison

Negotiations

Award

Purchase Orders

Deliveries

Budget / Commitments

Documents

Tasks

Activity

18.3 Overview

- Package identity, company/project/work package and responsible buyer/team.
- Current lifecycle stage and exception state from linked source records.
- Open demand, sourcing status, award/approval state, orders and critical delivery milestones.
- Permission-safe requested/committed/received/invoiced summary with source labels.
- Upcoming deadlines and recent package activity.

18.4 Allocation integrity

- Request quantities allocated to packages/RFQs/awards/POs use transactional controls to prevent over-allocation.
- Package links store authoritative record ID, source module, version where relevant and correlation ID.
- Archived Package history remains searchable subject to permissions.

19. Purchase Request, RFQ and PO Record Workspaces

19.1 Purchase Request page

Overview | Lines | Specifications / Documents | Project / Delivery Context | Revisions | Workflow | Tasks | Activity

19.2 RFQ page

Overview | Lines | Suppliers | Issued Revisions | Clarifications | Quotations | Comparison | Workflow | Documents | Activity

19.3 Purchase Order page

Overview | Lines | Commercial Terms | Delivery Schedule | Acknowledgment | Revisions | Receipts | Invoices / Payments (links) | Workflow | Documents | Activity

- Issued RFQ and PO content is immutable. Editable drafts and issued snapshots are separate.
- Record pages use the standard universal record pattern: identity, status, relationships, actions, documents and activity.
- Workflow card may show current stage, pending participants, decision history, lock state and submit/withdraw/resubmit actions without duplicating Workflow runtime.
- Finance/Inventory/Contracts links expose only authorized source fields.

20. Delivery, Supplier and Reporting Workspaces

20.1 Delivery page

Overview | PO / Schedule | Shipment / Arrival | Expected Quantity | Receipt Links | Exceptions | Returns / Claims | Documents | Tasks | Activity

20.2 Supplier management pages

Supplier Directory -> Supplier Page -> Qualification -> Documents -> Performance Scorecard -> Risk / Suspension

20.3 Reports and Analytics

- Spend, savings, cycle time, delivery, quality, supplier concentration, off-contract and project procurement are source-module analytics families.
- Every metric drills to source records and states its date/currency/company/project scope.
- Reporting must respect permission before aggregation and sensitive commercial fields.
- Heavy reports and exports run asynchronously and retain audit/correlation references.

20.4 Exact dashboard-to-report boundary

Dashboard stays operational

The Procurement Dashboard contains only the fixed operational regions defined in this PRD. Deep trend analysis, category strategy, long-period spend analysis and large data tables belong in Analytics/Reports, not added as dashboard widgets.

21. Approvals, Tasks, Calendar and Documents

21.1 Approvals / Workflow

- Approvals sidebar entry opens My Approvals / relevant decision queue inside Procurement shell.
- Procurement source pages embed workflow status but do not own routing, delegation, quorum or SLA logic.
- Workflow decisions always bind to exact source version; material source changes may invalidate/restart approval according to workflow policy.
- Assigned approvers act only within configured scope and cannot infer access to unrelated supplier/Finance/Contract content.

21.2 Tasks

- Procurement may create/link participant-based tasks for clarification, evaluation, delivery follow-up, supplier document renewal and exception resolution.
- Tasks owns task state, participants, due dates, checklists and collaboration.
- A team relationship does not automatically reveal confidential Procurement source records.

21.3 Calendar

- Upcoming may project RFQ deadlines, evaluation dates, expected deliveries, supplier reviews and procurement meetings.
- Calendar owns time projection; source modules retain record status.
- Meeting records/minutes remain in Meetings where applicable.

21.4 Documents / Transit

- All procurement attachments are controlled Document records with revision history and Document Passport.
- Documents may be referenced from supplier, request, package, RFQ, quotation, PO, delivery, return and claim without duplicating the underlying file.
- Transit supports working/draft collaboration before promotion to official controlled location when applicable.

22. Source-of-Truth and Cross-Module Integration

Domain	Owns	Procurement Dashboard uses
Procurement	Supplier purchasing profile; purchase demand; packages; RFQ/quotation/comparison/award; PO; delivery expectation; supplier performance/risk.	Authoritative procurement lifecycle and management projection.
Inventory	Product/SKU/unit; physical goods receipt; stock and posted inventory movements.	Product reference, receiving expectation and linked receipt summary.
Finance	Budgets, commitments/accounting references, AP invoices, payments and accounting.	Permission-safe budget/committed/invoiced/paid projections.
Contracts	Signed purchase/subcontract/framework agreements and legal terms.	Agreement/call-off reference; no copied legal contract state.
Assets	Long-lived asset activation, assignment, maintenance, depreciation context and disposal.	Draft Asset creation/reference from relevant procurement receipt.
Projects	Project hierarchy, phase/work package/cost code/site context.	Demand/package/delivery dimensions and filters.
Documents	Files and revisions.	All controlled attachments, supplier docs and issued artifacts.
Tasks	Execution work and collaboration.	Work Inbox/task links.
Workflow	Routing, participants, work items, decisions, SLA and escalation.	Approval status/work items and source-finalization requests.
Calendar / Notifications	Time projection / delivery channels.	Upcoming deadlines and notifications.

Permanent ownership rule

Procurement owns supplier purchasing context and the controlled journey from demand through sourcing, award, purchase commitment and delivery expectation. Connected modules retain authority for stock, accounting, legal agreements, asset lifecycle, projects, files, tasks, decisions and time. Cross-module records are linked through stable IDs/events, not copied into editable Procurement tables.

23. Authorization, Confidentiality and Separation of Duties

23.1 Procurement Manager authority

- Procurement Director / Manager may govern suppliers, requisitions, RFQs, evaluations, awards and approved PO issue within granted scope.
- Procurement authority cannot replace Finance, Inventory or Contracts ownership.
- Group Procurement scope must respect company-specific commercial confidentiality.
- Every dashboard card, count, row, field, activity event, export and deep link is backend permission filtered.

23.2 Protected content

Protected area	Examples
Competing bids	Quotation prices/terms before and during evaluation.
Evaluation	Technical/commercial notes, scoring inputs, conflict declarations.
Negotiation	Limits, internal target prices, negotiation notes.
Supplier sensitive	Bank change references, risk/conflict content, confidential documents.
Finance-linked	Budget/commitment/invoice/payment summaries.

23.3 Required SoD examples

- Prepare Award != Sole Award Approval.
- Create Supplier != Approve Supplier Bank Change.
- PO creator/issuer authority follows workflow and value/category policy; title alone does not grant final decision authority.
- A supplier evaluator with declared conflict is excluded/substituted/escalated according to policy.
- Emergency override initiator cannot independently close the post-event review.

24. Personal Dashboard Isolation and Shared-Route Context

24.1 Dashboard context key

tenant_id + company_id/group_scope + user_id + active_role_id + project/category/buyer filters + authorization_version

- Backend derives user identity from the authenticated session; never trust client-supplied user ID for dashboard ownership.
- Cache/read-model keys include tenant, active company/group, user/effective authorization, active role and applicable filters.
- Late responses from a previous company/project/role/filter context cannot overwrite the active dashboard.
- Access revocation invalidates relevant cached projections, saved session data and search/activity visibility.

24.2 Shared-route behavior

Opened content	Required shell behavior
Project	Authorized Project content opens inside Procurement shell.
Inventory Receipt / Product	Authorized Inventory content opens without switching to Inventory role.
Finance Budget / Invoice / Payment reference	Authorized Finance content opens permission-safe without switching to Finance role.
Contract / Asset	Authorized source record opens inside current Procurement shell.
Document / Task / Calendar	Shared page opens with Procurement sidebar/header preserved.
Approval	Workflow decision page opens with Procurement role context preserved.

Navigation rule

The authenticated active role/workspace resolves the shell. The route resolves page content. Permissions resolve visible data/actions. Opening a shared module must never silently switch the user to another role.

25. Platform Configuration and Saved Views

25.1 Configuration

- Every approved page, sidebar item, dashboard region, field group, status, workflow checkpoint, report and action is configuration-aware.
- Configuration may hide approved procurement features for a company/category/process but cannot create unapproved dashboard content.
- Disabled functionality disappears from navigation, search, APIs and layouts without empty cards or dead links.
- Configuration is evaluated separately from permission and cannot grant access or bypass lifecycle/workflow rules.
- Configuration changes are versioned/audited; historical records keep preserved values needed to render their issued state.

25.2 Saved views

- Procurement registers/workspace support private saved views with filters, grouping, sort, column selection and default/pinned state.
- Saved views never store inaccessible data and are revalidated after role/company/scope changes.
- Dashboard layout itself is fixed; saved views apply to registers/workspace, not dashboard rearrangement.

25.3 Search

- Global and module search must honor company/project/record/field permissions before returning results.
- Search can cover supplier, request, package, RFQ, quotation, PO, delivery and permitted linked records.
- Locked/safe metadata behavior follows platform search policy where configured; hidden commercial values never leak through snippets.

26. Frontend Architecture and State Management

26.1 Frontend composition

```
ProcurementShell
+-- ProcurementDashboardPage
|   +-- ScopeBar
|   +-- PrimaryCueGrid
|   +-- WorkInbox
|   +-- PipelinePanel
|   +-- DemandSourcingPanel
|   +-- OrderDeliveryPanel
|   +-- SupplierHealthPanel
|   +-- CommercialSummaryPanel
|   +-- UpcomingPanel
|   +-- RecentActivity
+-- ProcurementWorkspace
+-- Requests / Packages / RFQs / Comparison
+-- Orders / Delivery / Suppliers
+-- Reports / Settings / Shared Source Routes
```

26.2 State rules

- Server state is authoritative; client stores UI preferences, draft form state and reversible transition intent only.
- Query keys include tenant/company/project/category/buyer/filter/sort/pagination as applicable.
- Mutations invalidate exact affected aggregates and use event-driven incremental refresh where available.
- Long drafts may autosave with explicit saved state; issued records never autosave in-place modifications.
- Comparison matrix and large registers use virtualization.
- High-risk issue/award/PO/workflow actions are never treated as final until server confirmation.

27. API, Read Models, Events and Background Jobs

27.1 Dashboard and core API contracts

Contract	Purpose
GET /procurement/dashboard	Scoped KPI, work and dashboard summary.
GET /procurement/workspace	Board/table/timeline query.
GET /purchase-requests	Demand register.
GET /procurement-packages	Package register.
POST /rfqs/{id}/issue	Issue exact RFQ revision.
POST /comparisons/{id}/recommend	Create award recommendation.
POST /purchase-orders/{id}/issue	Issue immutable PO revision.
POST /deliveries/{id}/exceptions	Record delivery exception.
GET /procurement/reports/{reportKey}	Permission-safe report query/job.

27.2 API standards

- Cursor pagination for large lists/activity.
- Filter syntax supports company/project, status, date, supplier, category, value and saved-view state.
- Idempotency key is required for issue/conversion/commitment/receipt-link/bulk actions.
- If-Match or row_version is required for controlled updates/transitions.
- Errors include code, message, field, correlation ID, retryable flag and remediation hint.
- Bulk, export and heavy report operations are asynchronous with validation/error reporting.

27.3 Events / jobs

```

purchase_request.submitted
rfq.issued / rfq.response_received
quotation.received / quotation.revised
award.recommended / award.selected
purchase_order.issued / acknowledged / revised
delivery.scheduled / delayed / arrived
inventory_receipt.linked / service_receipt.accepted
supplier.performance_updated / document_expiring
supplier_claim.opened / resolved

```

- Dashboard summaries and supplier performance are precomputed/incrementally refreshed by events.
- Background jobs evaluate supplier expiry, RFQ/quote deadlines, unacknowledged POs, overdue deliveries, package risk and reconciliation.
- All jobs are idempotent, tenant-aware, correlation-aware and safe to retry; retries must not issue a second PO or duplicate Finance/Inventory effects.

28. Responsive UX, Accessibility and Required States

28.1 Responsive behavior

Surface	Behavior
Desktop	Wide virtualized tables, sticky identifiers, resizable columns, right-side preview drawers, horizontally scrollable comparison with frozen scope/totals.
Tablet	Two-column dashboard, collapsible navigation, touch-safe actions, bottom-sheet filters/actions and horizontal comparison/table scroll only when unavoidable.
Mobile Web	Compact cards, sticky record header, request/document capture, supplier/PO lookup, delivery arrival/shortage/damage/service receipt/return/claim actions; comparison becomes supplier-by-supplier cards.

28.2 Motion and accessibility

- WCAG 2.2 AA target with keyboard navigation, visible focus, semantic headings and accessible tables.
- Color is never the only status indicator.
- All drag actions have keyboard/menu alternatives.
- Charts include textual summaries and underlying data tables.
- Reduced-motion preference removes non-essential motion.
- High-risk commercial actions use restrained state feedback, not celebratory animation.

28.3 Required visual/data states

- Loading skeletons for dashboard/lists/comparison/activity.
- No requests / no RFQs / no POs / no deliveries / no suppliers / no work items.
- Partial dashboard failure with unaffected regions still usable.
- Permission denied without hidden-value leakage.
- Integration Pending / Failed / Reconciled for source-linked Finance/Inventory/Contracts/Assets states.
- Configuration disabled with clean reflow and no dead deep links.
- Concurrency conflict with Reload / Open New Version guidance.

29. Performance, Audit and Observability

29.1 Performance targets

Area	Target / behavior
Dashboard / standard list	Initial interaction target below 2 seconds under normal enterprise conditions.
Cached drill-down	Target below 250 ms where cached read model applies.
Common search	Target below 500 ms for permitted result sets under normal conditions.
Large data	Cursor pagination, virtualization, indexed dimensions and partitioning readiness for millions of lines/activity events.
Heavy jobs	Imports, exports, reports and large comparisons run asynchronously.

29.2 Audit / immutability

- Every lifecycle transition stores actor, timestamp, previous/new state, reason, source version and correlation ID.
- Every submitted request, issued RFQ and issued PO stores an immutable snapshot.
- Every quotation revision, comparison score/override, supplier qualification/status/risk change remains linked in append-only history.
- Imports, exports, bulk actions and configuration changes are logged.
- Issued commercial records are corrected only by revision, cancellation, replacement or controlled reversal-style mechanism.

29.3 Observability

- API latency/error/timeout/retry metrics by endpoint.
- Event outbox backlog, integration failures and reconciliation state.
- RFQ/PO issue success, idempotency collision and transition-conflict rates.
- Dashboard cache freshness, search indexing delay and supplier score calculation duration.
- Business telemetry may include cycle time, response rate, acknowledgment and delivery exception rate, but no hidden employee-surveillance score.

30. QA Strategy and Acceptance Criteria

30.1 Required test coverage

- Unit tests for dashboard aggregation, amounts/currency basis, due windows, pipeline mapping and permission gates.
- Integration tests across Projects, Inventory, Assets, Finance, Contracts, Documents, Tasks, Calendar, Notifications and Workflow.
- End-to-end supplier -> request -> package -> RFQ -> quotations -> comparison -> recommendation -> approval -> PO -> acknowledgment -> partial delivery -> receipt/exception -> return/claim.
- Negative authorization tests for tenant/company/project/record/field/action scope and commercial confidentiality.
- Concurrency/idempotency tests for award allocation, PO creation/issue, delivery acceptance context, returns and Workflow source finalization.
- Performance tests for large supplier directories, 10k-line requests, multi-supplier comparison, million-line history and concurrent dashboard use.
- Accessibility and visual regression tests across desktop/tablet/mobile, long labels, reduced motion and disabled-section reflow.

30.2 Acceptance criteria

1. Procurement login opens the exact Procurement shell and fixed Dashboard.
2. Primary cues are exactly the six defined cues and reconcile to source registers.
3. My Work shows only authorized assigned Procurement/Workflow/Task items.
4. Pipeline status comes from authoritative backend source state with no client-only lifecycle.
5. Demand/Sourcing/Ordering/Delivery/Supplier/Commercial panels reconcile with source records and scope.
6. Issued RFQs/POs and submitted requests are immutable snapshots; revisions remain linked.
7. Comparison and award preserve evaluation/override history and separation-of-duty behavior.
8. Workflow routing/decisions are shared projections; Procurement owns source finalization.
9. Inventory receipts and Finance invoice/payment states remain source-linked and cannot be edited from Procurement dashboard.
10. Supplier sensitive fields and competing bids are absent when permission is missing.
11. Shared source routes preserve Procurement shell and role context.
12. Platform-disabled elements reflow cleanly with no dead links.
13. Desktop/tablet/mobile and accessibility requirements pass.
14. Cross-user/company/tenant negative isolation tests pass.
15. No unapproved dashboard widget/sidebar item is present.

31. Explicitly Not Allowed and Definition of Done

31.1 Explicitly not allowed

- Additional Procurement dashboard KPI cards, charts or widgets not defined in this PRD.
- User rearrangement of dashboard regions in v1.
- Generic Finance, HR, Sales or executive company KPI widgets on the Procurement Dashboard.
- Automatic visibility of all group companies, projects, suppliers or commercial processes without scope.
- Duplicating Inventory receipts/stock, Finance invoices/payments, Contracts legal terms, Assets lifecycle, Projects, Documents, Tasks or Workflow records into editable Procurement dashboard storage.
- Editing issued RFQs/POs or silently replacing quotations/comparison history.
- Treating Workflow approval as the final Procurement state before source finalization succeeds.
- Hidden bid/negotiation/evaluator data leakage through counts, errors, search, export or activity.
- Client-only pipeline status, silent last-write-wins or duplicate issue actions.
- Supplier Portal, e-auctions, EDI/cXML/PEPPOL, OCR, AI sourcing/recommendation, live courier tracking, marketplace punchout or full offline sync in initial scope unless separately approved.
- Hard-coded title checks that replace platform capability/scope authorization.

31.2 Definition of Done

1. Exact Procurement sidebar, header and fixed dashboard layout are implemented.
2. Scope filters and six primary cues reconcile with authoritative source data.
3. My Work, pipeline, demand/sourcing, orders/delivery, supplier, commercial, upcoming and activity regions are fully connected.
4. Procurement Package, Request, RFQ, Comparison, PO, Delivery and Supplier workspaces deep-link correctly.
5. Workflow integration, source finalization, SoD and approval-matrix behavior pass end-to-end tests.
6. Cross-module Inventory/Finance/Contracts/Assets/Projects/Documents/Tasks links preserve ownership and permissions.
7. Role context, permission filtering and personal dashboard isolation pass shared-route/direct-link tests.
8. Responsive, accessibility, loading, empty, error and partial-failure states pass visual QA.
9. Performance/cache/idempotency/audit/security tests pass with representative enterprise data.
10. Dashboard scope lock is respected with zero unapproved elements.

Final implementation rule

If an element is not defined in PRD_Procurement_Dashboard, it does not appear on the Procurement Dashboard or Procurement role sidebar. New dashboard content requires an approved PRD revision.

Appendix A. Source Alignment Matrix

PRD decision	Primary source alignment
Procurement information architecture and dashboard KPI groups	PRD_Procurement_Module v1.0 - Information Architecture and Procurement Dashboard.
Procurement Manager role authority / protected commercial content	Complete Role & Permission Matrix - Procurement and Supplier Roles; protected fields; SoD.
Demand, packages, RFQs, quotations, comparison, awards, POs, delivery, suppliers	PRD_Procurement_Module v1.0.
Source ownership: Inventory / Finance / Contracts / Assets / Projects / Documents / Tasks	PRD_Procurement_Module architectural ownership and final rule + connected domain PRDs.
Workflow routing, matrices, participants, decisions, SLA and source finalization	Approvals & Workflow Engine v1.0.
Shared-module shell persistence	Role Context and Shared Module Navigation PRD.
Role Center, Work Inbox, Quick Create, responsive patterns, accessibility	Platform UI/UX Architecture and Premium Design System.
Exact six cues, fixed dashboard regions, exact sidebar and dashboard scope lock	PRD_Procurement_Dashboard v1.0 - new locked decisions.

Source conflict resolved

Procurement PRD vs Workflow PRD

The Procurement Module calls central approvals a future dependency because it predates the Workflow PRD. This dashboard uses the completed Workflow PRD for routing and decision orchestration while preserving Procurement as the source of truth for procurement business records and final status.

Appendix B. Complete Architecture and Route Context Map

```
/procurement/dashboard
-> My Work / Exceptions
-> Demand
  -> Purchase Requests / Replenishment / Emergency
-> Packages
  -> Procurement Package Workspace
-> Sourcing
  -> RFQs / Quotations / Comparison / Negotiation / Awards
-> Ordering
  -> Purchase Orders / Framework Call-Offs
-> Delivery
  -> Schedule / Expected Receipts / Service Receipts / Returns / Claims
-> Suppliers
  -> Directory / Qualification / Performance & Risk
-> Work
  -> Approvals / Tasks / Calendar / Documents
-> Reporting
  -> Reports / Analytics
-> Settings

SHARED SOURCE ROUTES
-> Project / Inventory / Finance / Contract / Asset / Document / Task / Workflow
-> active Procurement role shell remains resolved from session context
```

End-to-end business journey

```
Demand -> Request -> Package -> RFQ -> Supplier Response -> Comparison -> Negotiation -> Award
Recommendation -> Workflow Decision -> Procurement Source Finalization -> PO -> Supplier Acknowledgment ->
Delivery -> Inventory/Service Receipt -> Finance Invoice/Payment Links -> Completed
```

Architectural close

One platform experience. Procurement owns the purchasing lifecycle. Connected authoritative domains own their records. Workflow owns how decisions are routed. Every dashboard value is traceable to exact source records, exact scope and exact permissions.